

Sachin Koirala

 [cruvss](#) |  [sachinkoirala](#) |  +977 9804005727 |  sachinkoirala2016@gmail.com

PROFESSIONAL SUMMARY

Machine Learning Developer with over 1 year of experience in engineering end-to-end AI and data solutions. Skilled in computer vision, NLP, and multimodal models using Python and various ML frameworks to solve complex business solutions. Comfortable working with APIs, databases, and production systems, and experienced in leading small technical teams.

EXPERIENCE

ML Engineer, NextWave AI

July 2025 – Sept 2025

- Built an end-to-end ML pipeline for a horse race prediction system, including data collection, model training on 5 years of data, and deployment on Azure.
- Optimized the core logic of the ATS resume generator, reducing the average response time from 60 seconds to an average of 15 seconds (75% latency improvement).
- Designed and implemented a comprehensive EMIS database to manage all college-wide administrative data.
- Guided and trained interns on computer vision annotation workflows by establishing a hybrid cloud/local processing strategy, which reduced overall data processing time by at least 40%.

ML Intern (Team Lead), NextWave AI

May 2025 – June 2025

- Directed a baseball annotation project (20,000+ images) and implemented a YOLOv9 model for real-time pitch classification with 90%+ accuracy.
- Built and deployed FastAPI services for speech analysis, including fluency scoring, speech-to-text, and emotion detection.
- Supervised a 6-member team, delivering 3 ML projects on schedule.

ML Developer, Freelance

Jun 2024 – Dec 2024

- Developed an end-to-end glaucoma detection system using retinal fundus images, applying evolutionary algorithms for feature selection to improve diagnostic performance.
- Designed and implemented relational databases for multiple clients by translating business requirements into ER diagrams and optimized schemas.
- Built AI-powered educational tools using OpenAI APIs and Azure Text-to-Speech, including a class-wise content generator and personalized study support system..
- Authored technical documentation and research summaries on applied AI use cases, contributing to project clarity and stakeholder communication..

Technical Lead, Hult Prize at KhCE

Dec 2023 – Feb 2024

- Coordinated a 5-member tech team for a month-long event with 300+ participants.
- Oversaw Audio-Visual systems and digital platforms, ensuring zero downtime throughout the event.

TECHNICAL SKILLS

Programming Languages:	Python, C/C++, SQL, JavaScript, Bash
ML & AI:	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, LangChain, OpenAI API, RAG
Backend & Deployment:	FastAPI, REST APIs, Docker, Linux, Model Deployment
Data & Databases:	Pandas, NumPy, MySQL, PostgreSQL, MongoDB, SQLite
Tools:	Git, Postman, Streamlit, Gradio, Azure TTS

PROJECTS

Image Steganalysis using Ensemble Classifier

Repo

- Engineered an ensemble-based model to detect hidden messages in images with detection accuracy of 75%.
- Applied statistical feature extraction and classification methods for higher robustness.

Multimodal Malware Classification

Repo

- Developed a multimodal malware detection pipeline combining CNN based visual features with sequence modeling, improving resilience to code obfuscation.
- Research work accepted for conference publication; currently in copyediting stage.

101 Food Classification

Repo

- Trained a Swin Transformer on 100K+ images, achieving 85%+ accuracy through data augmentation and hyperparameter optimization.

LegalSathi

Repo

- Designed a chatbot using LLaMA 3 and LangChain to provide localized legal guidance in Nepal.
- Built standardized input schemas to improve response relevance for different demographics.

PUBLICATIONS

- Koirala et al. *Multimodal Approach for Malware Classification*. Status: Copyediting.

EDUCATION

Khwopa College of Engineering, Bhaktapur, Nepal	2021 – 2025
B.E. in Computer Engineering, Tribhuvan University	
Uniglobe Secondary School, Kathmandu, Nepal	2018 – 2020
Higher Secondary School, Science (NEB)	